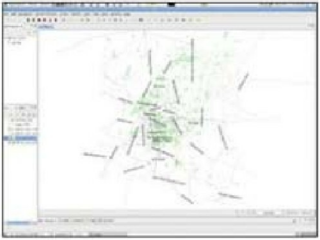

Figure 1: A map of Bangalore generated in uDig from Openstreetmap data

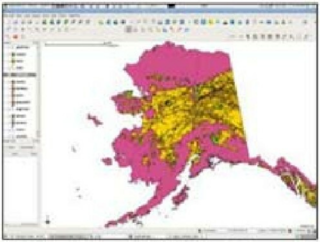

Figure 2: A map of Alaska generated in QGIS from the sample dataset

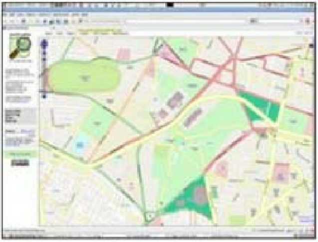

Figure 3: OpenLayers map of Bangalore city from 'Openstreetmaps.org

with your GRASS installation through the GRASS plug-in. This plug-in allows you to access, edit and run algorithms on the GRASS mapsets.

You can georeference an image—for instance, you can georeference a scanned city map by giving at least three reference points in the form of the latitudes, longitude.

QGIS displayed impressive performance while loading large spatial files and rendering maps.

OpenLayers

OpenLayers (OL) is a JavaScript library that allows you to put a dynamic map on a Web page. In short, it helps in building your own Google Maps-like applications. Openstreetmap.org and btis.in (Bangalore Transport Information System) use OL—that's how mainstream it is.

OL takes a map and/or vector data from a WMS or a WFS and displays it on a Web page with features like zoom, pan and layer-select, to list a few. The best thing is that it allows you to add a Google, Yahoo or Worldwind map as a layer so that you don't necessarily have to run your backend map service. Check out the gallery at *openlayers.org/dev/examples* to sample all that OL lets you build.

OL provides features to add markers, pop-ups, clustering, paging, draw geometries, etc.

Spatial databases

A spatial database supports the geometry datatype and has geometric functions like union, intersect, overlap, buffer, etc. A spatial database can answer a query like 'Find me all restaurants within 1 km of Forum Mall.'

PostGIS

PostGIS is an OGC (Open Geospatial Consortium) compliant tool that installs as a plug-in on the PostgreSQL object-relational database. The user needs to create a regular database in Postgres, and then run the *lwpostgis.sql* script provided in PostGIS to install geometry datatype support and spatial functions on the database. A table named *geometry_columns* maintains a record of the spatial tables, their respective geometry column, the geometry dimension, etc. PostGIS allows you to create a spatial index on the geometry column to speed up spatial queries.

The *pgRouting* plug-in provides the routing capability. It has implementations of Shortest Path Dijkstra, Shortest Path A*, Shortest Path Shooting Star and Travelling Sales Person algorithms.

As of now, PostGIS is the best open source option for a spatial database. A lot of GIS software use it as a data source. These include QGIS, uDig, GRASS, OpenJUMP, IGIS, GeoServer, MapServer and others.

SpatiaLite

SpatiaLite is an extension to SQLite, an SQL database engine. It conforms to OpenGIS specifications, and also